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November 2, 1972
SUPERSEDING
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FEDERAL SPECIFICATION

BUILDING PAPER, VEGETABLE FIBER OR INORGANIC FIBER REINFORCEMENT, REFLECTIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of reinforced, reflective building paper.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-320	- Pigment, Aluminum; Powder and Paste, for Paint.
UU-B-790	- Building Paper, Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant).
MMM-A-260	- Adhesive, Water-Resistant, (for Sealing Waterproofed Paper).
PPP-B-636	- Box, Fiberboard.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-P-1660	- Pallet, Expandable.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Fderal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-A-148 - Aluminum Foil.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-147 - Palletization Unit Loads on 40" x 48" Pallets.

FSC 5640

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D776 - Relative Stability of Paper (Effect of Heat on Folding Endurance).
D917 - Bleeding Resistance of Asphalted Paper at Elevated Temperature.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical Association of Pulp and Paper Industry (TAPPI) Standards:

T403 - Bursting Strength of Paper.
T404 - Tensile Breaking Strength of Paper and Paperboard.
T410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T433 - Water Resistance of Paper and Paperboard (Dry-Indicator Method).
T452 - Brightness of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The reflective building paper shall consist of reinforced single sheet or laminated kraft paper having a continuous aluminum reflective surfacing on one or both sides, as specified (see 6.2).

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Paper. Paper shall conform to UU-B-790, type I, grade A.

3.2.2 Aluminum foil. Aluminum foil shall conform to MIL-A-148.

3.2.3 Aluminum paste and powder. Aluminum paste and powder shall be of the leafing flake type as specified in TT-P-320.

3.2.4 Adhesive. Adhesive used in bonding shall be as specified in MMM-A-260, type I or II.

3.3 Physical requirements. The reflective building paper shall have a basis weight of 112 pounds maximum per 1000 square feet and shall have either a dry tensile strength of 28 psi minimum or a dry bursting strength of 84 psi minimum for two sides reflective paper and 28 psi minimum for one side reflective paper. In addition, building paper shall be capable of conforming to the capability requirements of table I, (see 6.4).

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TABLE I. Capability requirements 1/

Requirement	Value
Brightness, minimum degree.	25%
Water resistance, minimum.	12 hours
Resistance to bleeding.	No evidence of bleeding except that bleeding at the cut edge may be disregarded.
Pliability.	No evidence of cracking or breaking.
Mildew resistance.	No evidence of mildew growth.

1/ See 4.3.3

3.4 Bond of reflective surface to backing. The reflective building paper shall conform to the bonding test specified in 4.4.2.2 with no evidence of curling, separation of reflective surface from backing, or separation or loosening of the reflective surfaces when lightly rubbed while wet or after being dried.

3.5 Fabrication. The paper shall be coated with a reflective surface. The reflective surface shall be aluminum foil, aluminum paste, or aluminum powder. The edges of the paper shall be reinforced so as to resist tearing when nailed or stapled. When aluminum foil is used there shall be no more than one splice of the reflective surface on each side of the finished roll.

3.6 Dimensions. Unless otherwise specified (see 6.2) the building paper shall be furnished in rolled continuous sheets, 36 inches wide plus or minus 1/2 inch, by 250 feet in length plus or minus 12 inches.

3.7 Identification marking. When specified (see 6.2), each roll of building paper shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. The reflective material shall have flat uniform surfaces and shall not be crimped, torn or punctured. The finished building paper shall not stick together so as to cause tearing or damage to the reflective surface when unrolled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification inspection. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Examination of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, level S-3.

4.3.1.1 Lot. For the purpose of inspection, a lot shall consist of all the rolls of paper of the same material and manufacture offered for delivery at one time.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 and 4.4.2.2 except that tests for capability requirements, (see Table I), shall be conducted as required. The AQL for all tests shall be 4.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The building paper shall be examined for the defects listed in table II.

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TABLE II. Classification of defects

Examine	Defect
Materials.	Not as specified (3.2).
Fabrication	Edges not reinforced (3.5). Reflective surface splice not as specified (3.5).
Dimensions	Not as specified (3.6).
Marking	Not as specified (3.7).
Workmanship	Reflective surfaces not flat and uniform (3.8). Reflective surfaces torn or punctured (3.8). Tearing or damage to reflective surface when unrolled (3.8).

4.4.2 Tests.

4.4.2.1 Physical. The building paper shall be tested in accordance with the methods listed in table III and as specified herein. Nonconformance to the requirements of 3.3 shall constitute failure of these tests.

TABLE III. Physical tests

Test	Specification	Method No.
Basis weight	TAPPI	T 410
Bursting strength (dry)	TAPPI	T 403
Tensile strength (dry)	TAPPI	T 404
Brightness 1/	TAPPI	T 452
Water resistance 1/	TAPPI	T 433
Resistance to bleeding 1/	ASTM D917	----
Pliability 1/	ASTM D776	----
Mildew resistance 1/	Fed. Std. No. 191	5750

1/ Capability requirements (see 4.3.3).

4.4.2.2 Bond of reflective surface to backing. Five specimens, 2 by 4 inches, from each sample roll, shall be immersed in water at 20°C. for 6 hours. After the immersion period remove the specimens from the water and examine. Nonconformance to 3.4 shall constitute failure of this test.

4.5 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, workmanship, and contents comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The inspection level shall be S-2 and the AQL shall be 2.5 expressed as defects per 100 units.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, level S-2.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the defects specified in table IV. AQL shall be 2.5 percent defective.

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TABLE IV. Examination of preparation for delivery

Examine	Defects
Marking (interior and exterior containers)	Incorrect; incomplete; illegible; omitted; of improper size, location sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling when required.
Contents	Number of rolls per shipping container, more or less than required.
Containers	Not as specified, not close fitting for level A or B.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each roll of building paper, shall be packaged in a close-fitting fiberboard box conforming to PPP-B-636, class weather resistant, grade as applicable to the weight limitation of the special requirements in table II of PPP-B-636, style optional. Closure shall be as specified for class weather resistant boxes in the appendix PPP-B-636.

5.1.2 Level B. Each roll of building paper shall be packaged in a close fitting fiberboard box conforming to PPP-B-636, class domestic, grade 275, style optional. Closure shall be in accordance with the requirements of Method I for class domestic boxes as specified in appendix to PPP-B-636.

5.1.3 Level C. Each roll of building paper shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The rolls of building paper, packaged as specified in 5.1.1. in the quantities as specified (see 6.2) shall be packed in a crate conforming to PPP-C-650, type V, style A, and the contents protected with a full shroud in accordance with the appendix to PPP-P-650.

5.2.2 Level B. The rolls of building paper, packaged as specified in 5.1.2, do not require additional packing, but shall be shipped on a pallet as specified in 5.4.

5.2.3 Level C. The rolls of building paper, in quantities as specified (see 6.2) shall be packed so that they are acceptable to the carrier and shall insure safe delivery to the receiving activity. Shipping containers shall be in accordance with the Uniform Freight Classification rules or the National Motor Freight Classification rules as applicable.

5.3 Marking. In addition to any other markings specified in the procurement document, the building paper rolls shall be marked in accordance with Fed. Std. No. 123 for civil agency shipments or MIL-STD-129 for military agency shipments as applicable.

5.4 Palletization. As specified in 5.2.2, level B packaged rolls of building paper, shall be palletized on pallets conforming to PPP-P-1660, of the type suitable for the quantity and weight to be shipped per pallet. The boxed rolls shall be palletized in accordance with figure 18, type XVII, and strapped in accordance with paragraph 5.1.2.2 of MIL-STD-147.

6. NOTES

6.1 Intended use. The building paper is intended for use as insulation and vapor-barrier material in prefabricated building construction.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Whether the paper shall have reflective surface on one or both sides (see 3.1).
(c) Length and width of roll required if other than as specified (see 3.6).
(d) When marking in accordance with MIL-STD-130 is required (see 3.7).
(e) Level of packaging and level of packing required (see 5.1 and 5.2). Level B packaging (see 5.1.2 and 6.3) is intended to provide limited but economical protection and shall be specified only when it is determined the building paper will be held in covered storage for an indefinite period.
(f) Quantities of rolls per crate (see 5.2.1).

6.3 Level B packaging. Level B packaging (see 5.1.2) has been developed for use by the Army and Air Force for specific conditions specified in 6.2. For Navy procurements, details of level B packaging, when required, should be developed at the time of procurement when all conditions are known.

6.4 Capability requirements. The term "capable of" as used in this specification means that the test need not be performed by the producer of the building paper. However, should subsequent testing by the procuring activity at the time of delivery establish that the material does not meet these requirements, the building paper may be rejected.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force 82

Review activities:

Army - ME
Air Force - 82

User activity:

Army - CE

Preparing activity:

GSA-FSS

Military Coordinating Activity:

Project No. 5640-0260

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